

Cooling cravers to the carefree: four archetypes of thermal preference from city-scale longitudinal smartwatch micro-surveys

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Abstract

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Keywords: thermal comfort, thermal preference, personal comfort, wearable sensing, ecological momentary assessment, behavioural archetypes

1. Introduction

1.1. Thermal preference varies between people under similar conditions

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1.2. Repeated in-situ data can reveal behavioral response patterns

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2. Methods

2.1. Study design, cohort, and data source

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Figure 1: Cozie smartwatch micro-survey interface. The thermal-preference question offers Cooler, No change, and Warmer responses; the accompanying location question records the space category at the time of response.

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2.2. Survey measures, outdoor heat, and environmental-control context

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2.3. Participant eligibility and person-level features

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2.4. Archetype discovery and interpretation

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2.5. Mechanistic scrutiny and robustness checks

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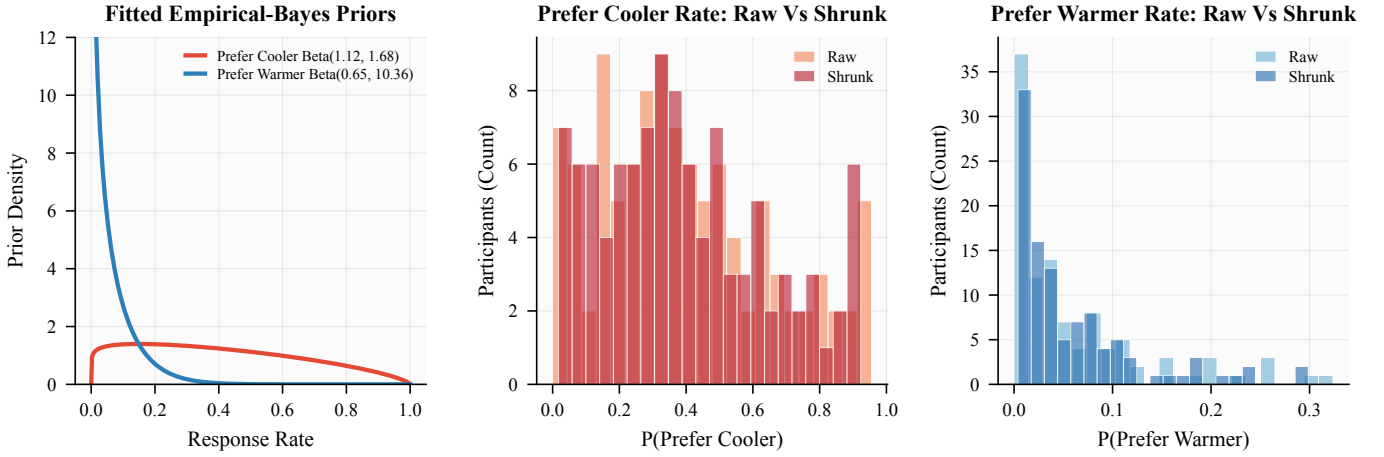


Figure 2: Empirical-Bayes feature construction. The left panel shows fitted Beta priors for prefer-cooler and prefer-warmer response rates; the center and right panels compare participants’ raw and shrunk rates for the two features.

3. Results

3.1. Cohort coverage and thermal-preference context

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3.2. Behavioral thermal archetypes

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3.3. Heat dose-response and environmental-control context

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3.4. Robustness and boundary membership

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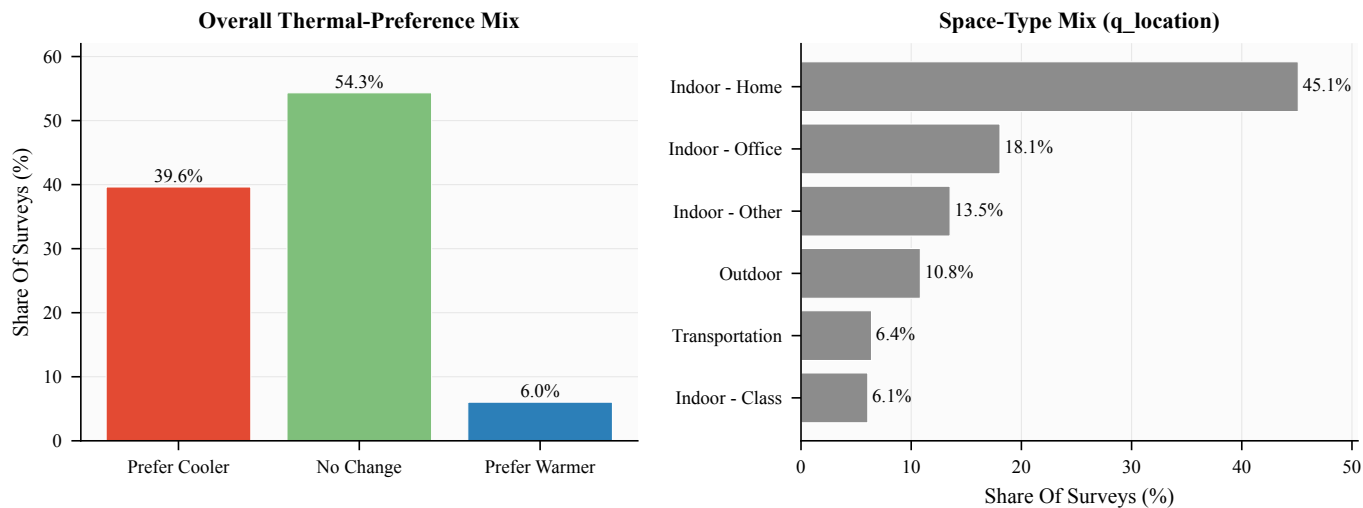


Figure 3: Survey composition. The left panel shows the overall thermal-preference mix and the right panel shows the distribution of survey responses across recorded space types.

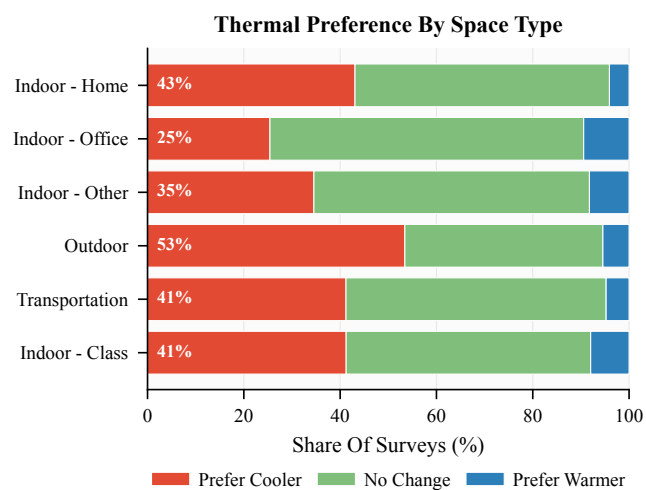


Figure 4: Thermal-preference responses by recorded space type. Stacked bars show the share of surveys indicating Prefer Cooler, No Change, or Prefer Warmer within each space type.

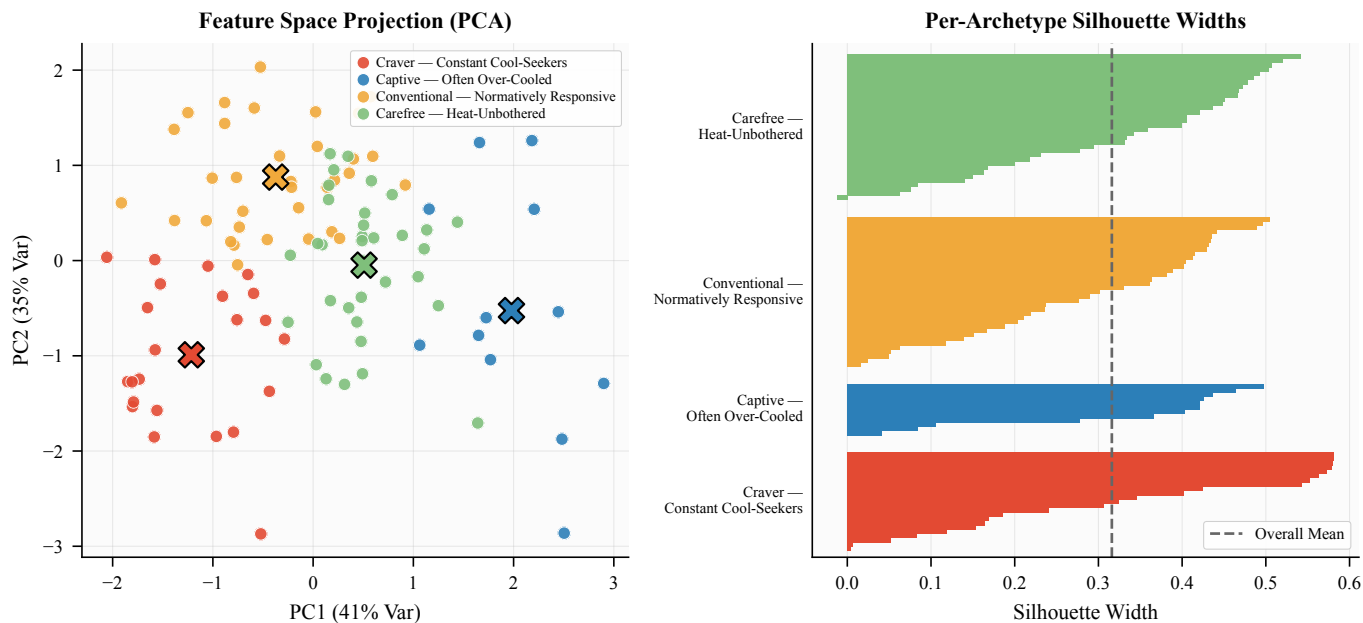


Figure 5: Four-archetype clustering diagnostics. The left panel projects the standardized feature space onto its first two principal components; the right panel shows silhouette widths for individual participants, grouped by archetype.

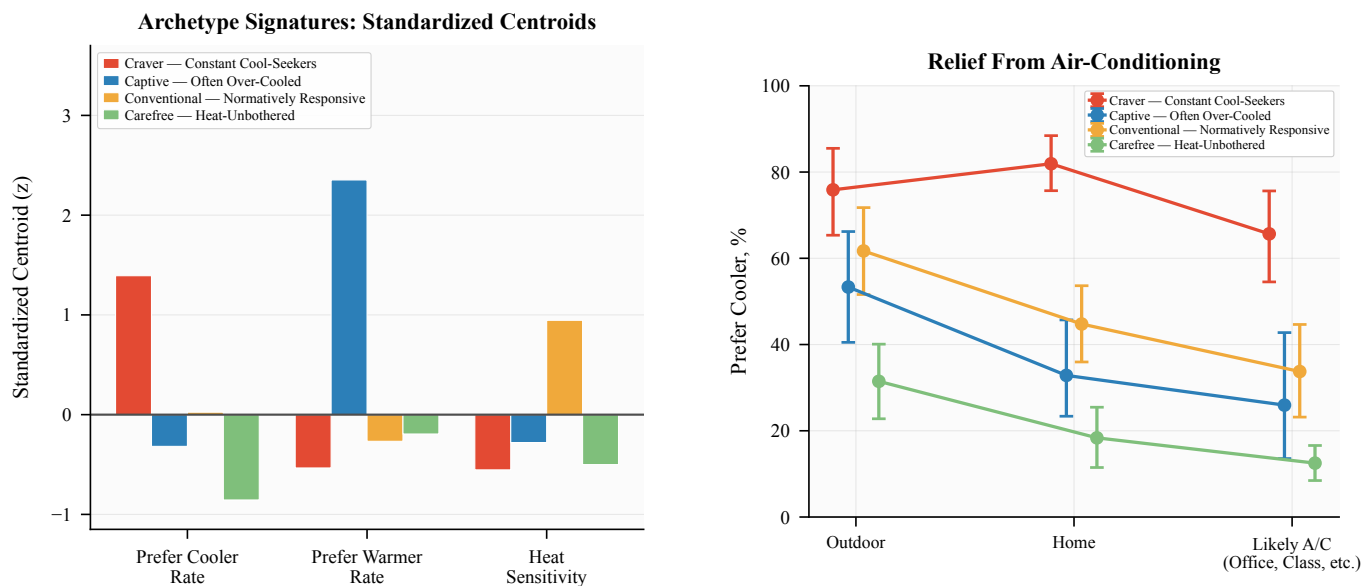


Figure 6: Standardized feature signatures of the four behavioral thermal archetypes. Lines show cluster centroids for prefer-cooler rate, prefer-warmer rate, and heat sensitivity.

Figure 7: Prefer-cooler responses across outdoor, home, and likely air-conditioned contexts for each archetype.

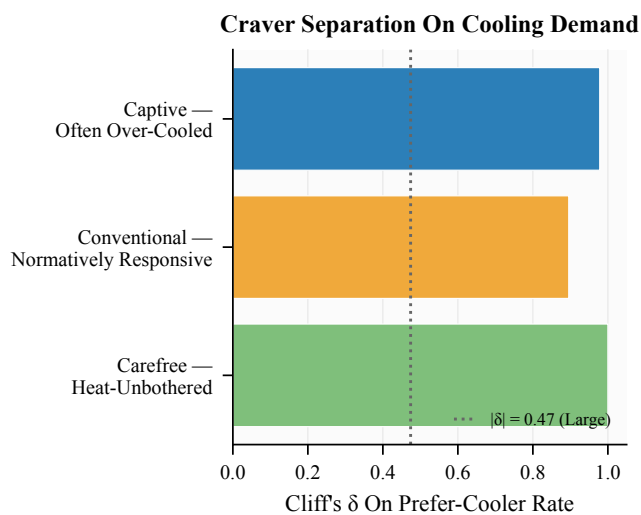


Figure 8: Pairwise separation of the Craver archetype from the remaining archetypes on participants' prefer-cooler rates, summarized using Cliff's delta.

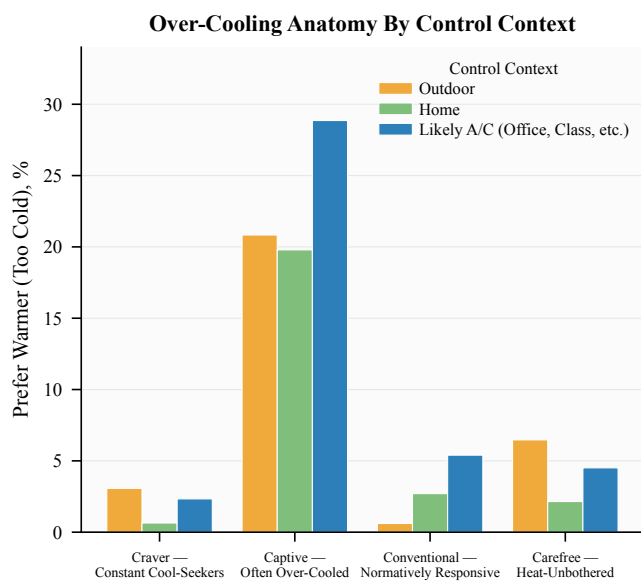


Figure 9: Prefer-warmer responses by archetype and control context. Points show the percentage of responses indicating a preference for warmer conditions in outdoor, home, and likely air-conditioned contexts.

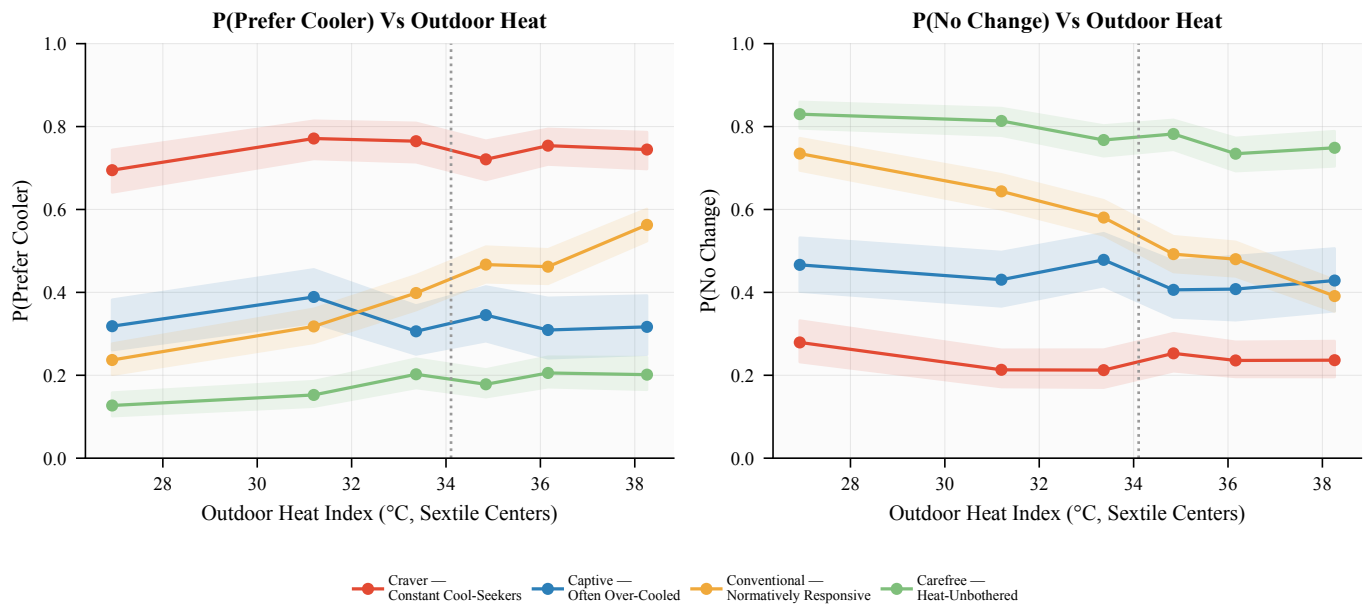


Figure 10: Thermal-preference dose response to outdoor heat. The top panel shows the probability of preferring cooler conditions and the bottom panel shows the probability of no change across outdoor heat-index sextile centers for each archetype.

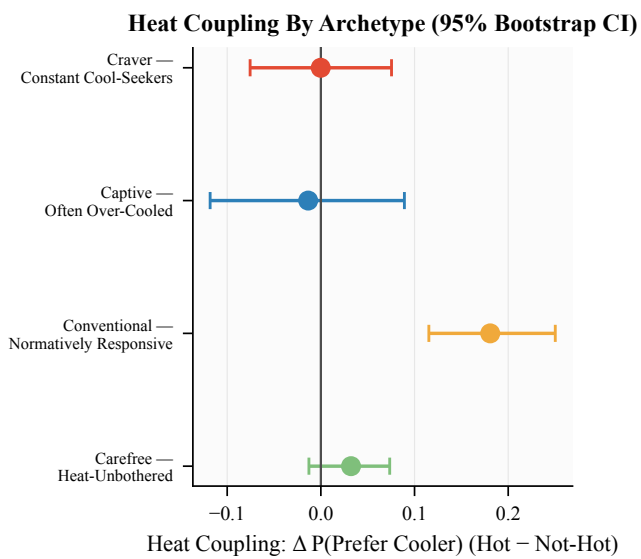


Figure 11: Heat coupling by archetype. Points and horizontal intervals show the difference in participants' prefer-cooler probability between hot and not-hot observations, with 95% bootstrap confidence intervals.

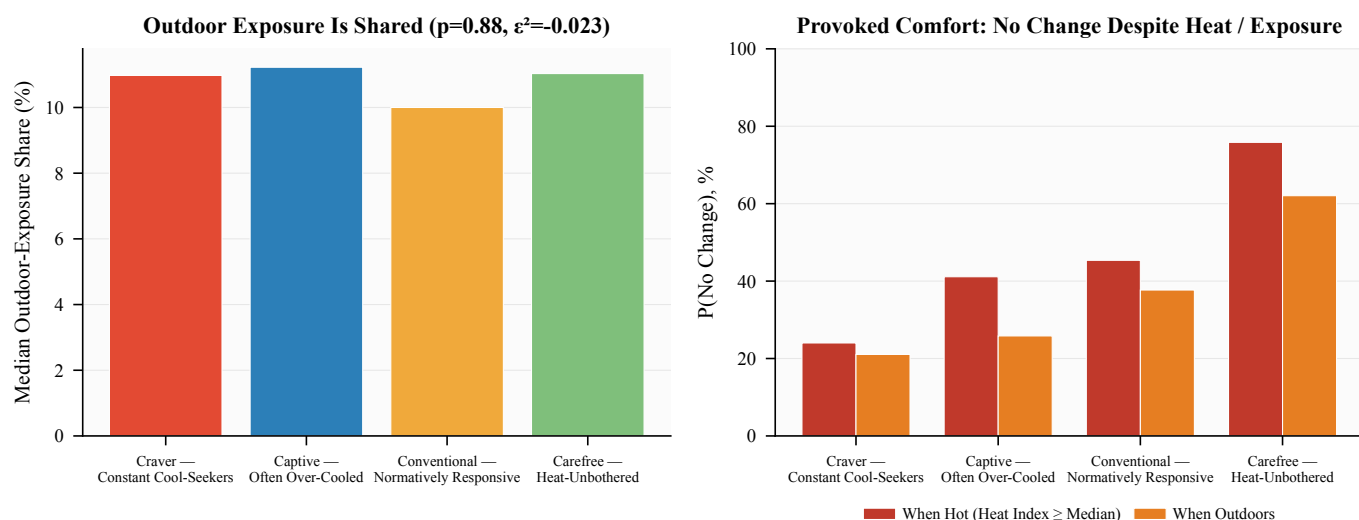


Figure 12: Outdoor exposure and provoked comfort by archetype. The left panel summarizes participants' outdoor-exposure share; the right panel shows the probability of reporting No Change when observations are both hot and outdoors.

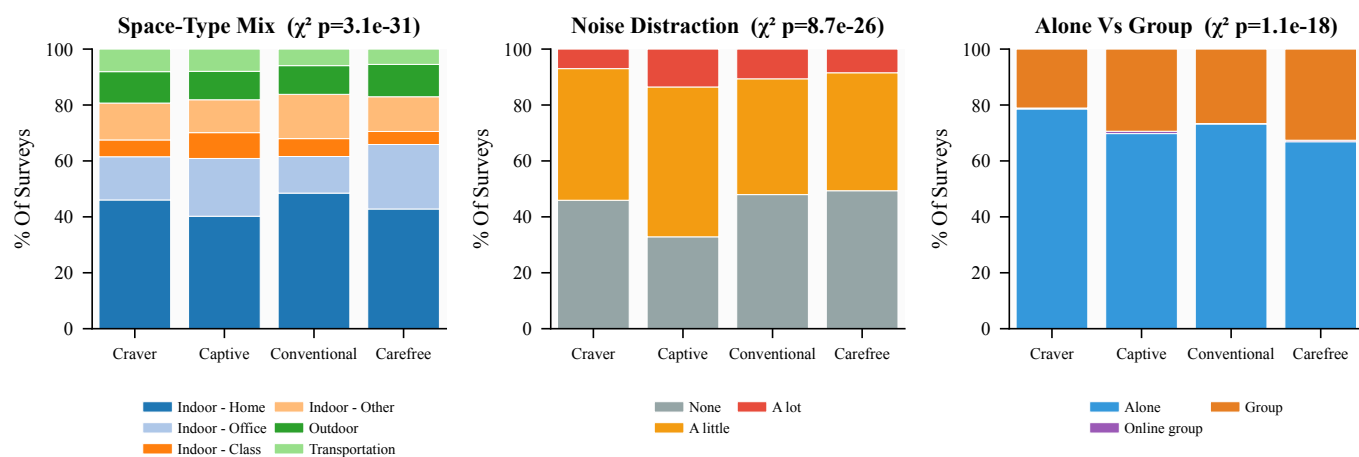


Figure 13: Survey characteristics not used to construct the archetypes. From left to right, panels show the distribution of space type, noise distraction, and alone-versus-group context across the four archetypes.

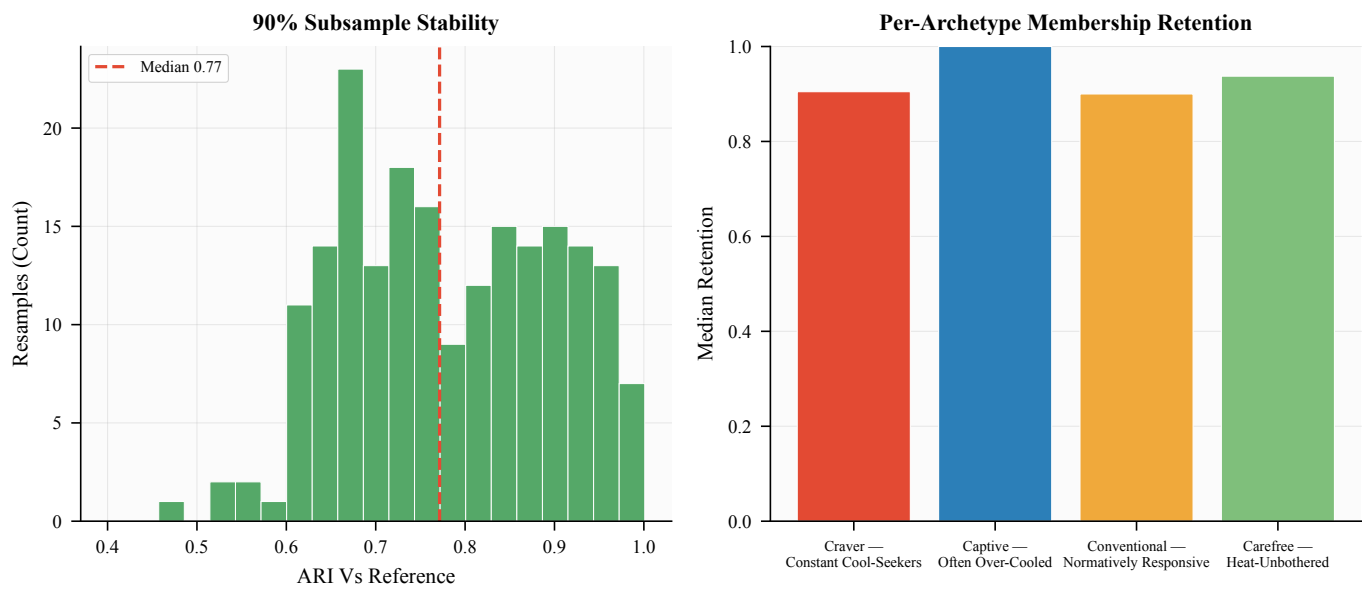


Figure 14: Clustering stability under 90% participant subsampling. The left panel shows adjusted Rand index values relative to the reference clustering; the right panel shows median archetype-membership retention.

4. Discussion

4.1. What the archetypes add to thermal-comfort research

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4.2. Implications for buildings, controls, and personalization

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4.3. Strengths and limitations

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4.4. Future work

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5. Conclusions

CRedit authorship contribution statement

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CRedit authorship contribution statement

CRedit roles to be finalized by the authors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Data availability

The data and analysis materials will be made available in the associated public repository, subject to the privacy protections described in the accompanying data documentation.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used generative AI tools to assist with organizing and polishing manuscript source files. The authors reviewed and edited all content and take full responsibility for the final manuscript.

Appendix A. Supplementary methods and robustness detail

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